

Identification of Temperature-Dependent Water Quality Changes during a Deep Well Injection Experiment in a Pyritic Aquifer

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Artificial recharge is a technique used increasingly to supplement drinking water supplies. To assess the potential water quality changes that occur during subsurface passage, a comprehensive deep-well injection experiment was carried out for a recharge scheme, where pretreated, aerobic surface water was injected at 300 m depth into an anaerobic aquifer. Water quality parameters were recorded over the 854-days long injection phase. The evolution of the major ion and redox chemistry was analyzed with a three-dimensional reactive multicomponent transport model. It was found that the oxidation of pyrite was the main driver for water quality changes and that reaction rates depended significantly on the spatially/temporally varying groundwater temperature. With the temperature-dependency of the oxidation reactions incorporated into the model, the simulations give an accurate picture of the temporal and spatial evolution of the hydrochemical changes that occurred during the experiment. To delineate the influence of physical and chemical processes on local concentration changes the results of the reactive transport model simulations were compared with the corresponding results from nonreactive simulations. The study emphasizes the suitability of mechanistic multicomponent reactive transport modeling as an integrative tool for data analysis when physical transport and chemical processes interact.

Introduction

Artificial recharge and aquifer storage and recovery (ASR) are increasingly used to supplement drinking water supplies (1). In the case of deep well injection and ASR, water of different quality is directly introduced into a relatively deep, well-protected and normally anoxic aquifer. Very often these deep aquifers contain considerable amounts of pyrite, which is oxidized and dissolves under the influence of the oxic

injectant. In many of these cases the pyrite oxidation process is the key reaction with respect to water quality changes during soil passage (2). Possible adverse impacts of pyrite oxidation on the evolution of the water quality include the undesirable release of dissolved iron and trace elements such as As, Co, Ni and Zn (3–5) and acid buffering reactions, which raise either the acidity or hardness. On the other hand, minerals such as ferrihydrite or goethite, which freshly precipitate in the oxidized zone near the injection wells may act as important surfaces for (trace) metal sorption (2, 6) and for the attachment and subsequent decay of pathogenic viruses (7, 8). For a better understanding of how physical, chemical and biological processes interact in pyritic aquifers during artificial recharge, it is important to establish a quantitative framework that integrates all factors which, at the field-scale, potentially affect pyrite oxidation and associated secondary reactions. The capability to quantify the reactivity of natural reductants such as pyrite is not only significant for artificially recharged systems, benefits are also expected for the application of stimulated in situ bioremediation measures of contaminated aquifers (9–12), for treatment of nitrate contaminated water by aquifer passage, and for the prediction of groundwater quality in areas affected by, for instance, lowered groundwater tables and excessive manure or fertilizer applications.

Numerical modeling, in particular geochemical transport modeling, has shown to be a useful tool for a comprehensive interpretation of laboratory and field-scale observations of pyrite oxidation. For example, Postma et al. (13) and Engesgaard and Kipp (14) presented equilibrium based modeling studies on nitrate removal by pyrite oxidation at the Rabis Creek aquifer in Denmark. Furthermore, Appelo et al. (15) modeled pyrite oxidation by H_2O_2 in a laboratory column and demonstrated the importance of various buffering reactions. Eckert and Appelo (12) reported a modeling study where they quantified the effect of pyrite oxidation during a field-scale BTEX remediation experiment. None of these studies was affected by a significant temporal or spatial variation of water temperatures. In contrast, the injection of surface waters, e.g., from surface water storage ponds, can lead to considerable seasonal temperature fluctuations within the aquifer. Recent studies by Stuyfzand (2) and Stuyfzand et al. (16) indicate that pyrite oxidation rates can vary strongly, depending on (i) the temperature of the injected water and (ii) the competition between pyrite and bulk organic material in the aquifer (BOM (11)). Both factors are thought to strongly influence the evolution of the redox chemistry and acidity within the target aquifer during artificial recharge and so impact the mobility of many contaminants.

Incorporating the temperature dependence of the key redox reactions, the present study provides a detailed quantitative description of the flow and physical transport processes, combined with the temporally and spatially varying major ion and redox chemistry that was observed during the deep well injection experiment at the field site *Dizon* in The Netherlands.

Material and Methods

Field Site and Investigation Program. Two Dutch water supply companies (Brabant Water and WML) investigated the technical feasibility for deep well recharge of canal water to combat the drawdown of water tables and to restore wetlands in selected areas. A pilot plant was built along the bank of the canal Zuid-Willemsvaart near Someren in Southern Netherlands, comprising an intake, a pretreatment facility, an injection well, 4 observation wells, a 98 m distant

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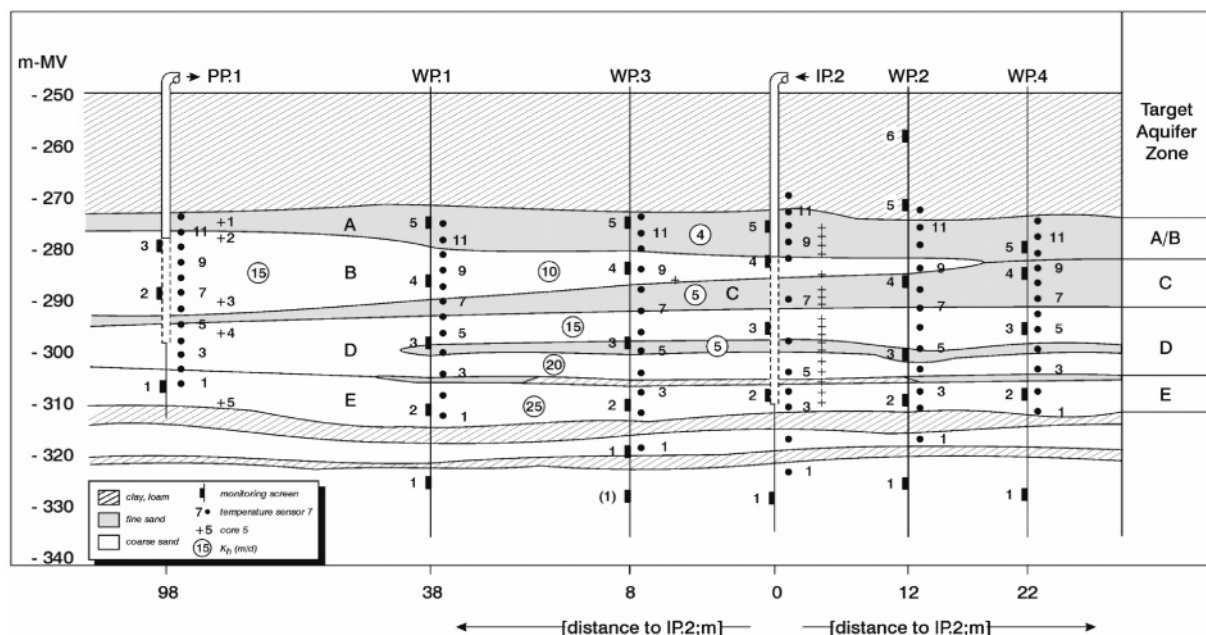


FIGURE 1. Position of the injection well (IP.2), recovery well (PP.1) and monitoring wells (WP.1–4) in cross section, showing their well screens, temperature sensors, the stratification within the target aquifer, estimated permeability distribution (before model calibration, K_h in m/d) and aquifer cores for geochemical analysis. A–E = hydrological and chemical zones within the target aquifer. M–MV = meters below land surface.

pumping well, and a discharge of the recovered water to a storage pond which drained the water back to the canal (Figure S1). After various tests with the pretreatment facility, the large-scale deep well experiment started on July 8, 1996 and ended in May 1999. The injection by one well and pumping by the 98 m distant well operated without significant interruptions. A detailed site characterization and monitoring program accompanied the trial injection. The location of the wells and the position of the sampling ports are shown in Figure 1. Temperature measurements were conducted on the infiltration water and groundwater using a thermistor (precision 0.01 °C). In each well the deepest piezometer (which was not pumped for sampling) was used to log temperature-depth profiles at the depth of the target aquifer. The diameter of all piezometers was 5 cm, which is small enough to minimize convective flow in the standing water column due to vertical temperature gradients (17).

Groundwater samples were obtained by suction lift from PVC-piezometers (well screen 2 m long). The water content of riser plus well screen was purged 3 times prior to sampling. EC, pH, alkalinity and O_2 were measured on site. The portion to be analyzed for most trace elements, Ca^{2+} , Mg^{2+} , Fe, Mn, SiO_2 and PO_4 , was filtrated in situ over 0.45 μm , directly acidified to pH = 1.5 using HNO_3 -suprapure and was stored in the dark at 4 °C in polyethylene vessels rinsed out with HNO_3 and the water to be sampled. Main constituents were analyzed with conventional, well-standardized analytical methods, and most trace elements were analyzed using Atomic Absorption Spectrophotometry (AAS) with graphite furnace. Sediment cores from the target aquifer were obtained using a core catcher during drilling of the monitoring wells. The sediment cores including their pore water were immediately sealed on site using liquidized paraffin and kept at 4 °C in the dark till analysis. Samples were opened and pretreated in an anoxic glovebox. The samples were analyzed for, among others, total content of 27 elements by X-ray fluorescence, organic matter and calcium carbonate by HCl-extraction of Ca^{2+} (correcting for exchangeable Ca; ppm level), organic carbon by pyrolysis, exchangeable cations using $SrCl_2$, sulfide species (especially pyrite) with a sequential extraction, non-Si-bound trace elements by aqua regia extraction and

amorphous Al-, Fe- and Mn-(hydr)oxides by ammonium oxalate/oxalic acid.

Numerical Model. The MODFLOW/MT3DMS-based reactive multicomponent transport model PHT3D (18, 19) was used for the transport simulations of the trial injection at the Dizon site. PHT3D couples the three-dimensional transport simulator MT3DMS (20) with the geochemical model PHREEQC-2 (21) through a sequential operator-splitting technique (22–25). The two individual models are widely used in the groundwater/geochemistry community and were individually thoroughly tested and documented. The coupled model has been successfully benchmarked (18, 19) and applied for a wide range of reactive transport problems (19, 25–27).

Conceptual Model, Model Discretization and Groundwater Flow Model. The deep well injection and extraction scheme creates a fully three-dimensional flow field that strongly determines the hydraulic gradients and mass transport in the vicinity of the scheme. The regional groundwater flow represented a negligible component of the flow field. The symmetrical flow-field that resulted from the injection of 720 $m^3 day^{-1}$ (between day 0 and 726) and 960 $m^3 day^{-1}$ (between day 727 and day 854) could be approximated by a half-model of 263 m length (in flow direction) and 124 m width (perpendicular to the main flow direction and symmetry axis, see Figure S2). In the vertical direction the model was discretized into 12 layers of different conductivities between –273 m and –340 m depth, as proposed in the conceptual model (28). The measured hydrochemical data from below and above the target aquifer indicated that no vertical fluxes occurred through the top and bottom of the (modeled) aquifer. A no-flow boundary was assumed at the model boundaries parallel to the main flow direction. One of these two boundaries is the symmetry plane and the other boundary is a plane in 124 m distance. A fixed head boundary was assumed for the remaining two boundaries, allowing water and mass to be exchanged through the boundaries. For the initial model runs the hydraulic conductivity distribution was identical with the hydraulic conductivities proposed by Stuyfzand (28). It was then slightly modified during the course of the model

calibration, as discussed below. Since the site characterization focused on a single transect along the axis defined by the injection and extraction well, no data were available to support a heterogeneous conductivity distribution within the model perpendicular to the main transect. The discretization of the model and the selected boundary conditions are shown schematically in Figure S2.

Nonreactive Transport Model. Chloride was found to be a suitable tracer for the calibration of nonreactive transport due to the clear concentration contrast between background ($2.54 \times 10^{-4} \text{ mol L}^{-1}$) and injection water ($1.56\text{--}3.24 \times 10^{-3} \text{ mol L}^{-1}$). Based on the three-dimensional flow model, an advective-dispersive transport model for chloride was set up. To account for the variability of the chloride concentrations in the injection water, the total simulation time of 854 days was discretized into 39 stress periods.

The observation data from up to 5 different screens (varying in depth) from observation wells WP.1-WP.4 and the measured chloride concentration at the pumping well (PP.1) were used to modify (i) layer-wise the initial conductivity distribution, (ii) the horizontal conductivity of the boundary grid-cells which regulates the fluxes that can occur through the model boundaries and (iii) the longitudinal and transversal dispersivities.

Heat Transfer. During the whole trial period the temperature of both the infiltration water and the groundwater was recorded at various depths in all available monitoring wells. While the temperature of the (deep) ambient groundwater remains rather constant at approximately 17°C , the temperature of the injectant fluctuated in a seasonal pattern between a minimum of 2°C and a maximum of 23°C . It was observed that temperature differences at the Dizon site propagate at approximately half the velocity of the chloride concentration changes. A subset of the recorded temperature data served as constraints for a heat transport model that was aimed at improving the calibration of the hydraulic conductivity distribution and, more importantly, to provide a good representation of the spatial and temporal variability of the temperature field to be used for the computation of the reaction rates of temperature-dependent chemical reactions.

Heat transfer in aquifers is generally governed through (i) heat transport by the fluid phase (ii) conductive transport of heat in the aquifer matrix, and (iii) heat exchange between the aqueous phase and the aquifer matrix, depending on the temperature gradient (29). To include heat transfer into the present reactive transport model of the deep well injection experiment, a simplified model formulation that neglects the conductive transport of heat within the aquifer matrix was employed. Using this simplification, heat transport can be modeled as analogous to the transport of a solute that undergoes sorption (21)

$$R_T \frac{\partial T}{\partial t} = -v \frac{\partial T}{\partial x} + \kappa_L \frac{\partial^2 T}{\partial x^2} \quad (1)$$

with the retardation factor

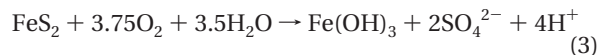
$$R_T = 1 + \frac{(1 - \theta) \rho_s k_s}{\theta \rho_w k_w} \quad (2)$$

where T is the temperature ($^\circ\text{C}$), θ is the porosity of the aquifer matrix, ρ_w and ρ_s are the density of water and the solid phase (aquifer matrix), respectively, k_w and k_s are the specific heat of water and the solid phase, respectively, and κ_L is the thermal dispersion coefficient. The latter represents both, pure diffusion and dispersion resulting from advective flow (21).

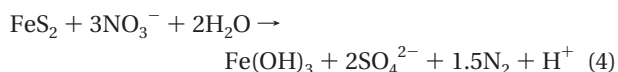
Reaction Network. For the reactive transport simulations, the reaction network includes the equilibrium-based

speciation and redox reactions of all major ions, cation exchange and mineral equilibrium for ferrihydrite ($\text{Fe}(\text{OH})_3$). Sediment-bound (immobile) organic matter (BOM) was considered as a separate, immobile phase similar to a mineral phase ($\log K_{\text{BOM}} = -4.778$) to provide a continuous source of DOC. Its oxidation was not explicitly modeled.

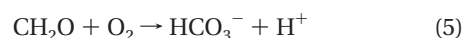
In addition to the equilibrium reactions, kinetic reactions were formulated and included for the key reactions, i.e., pyrite oxidation by oxygen



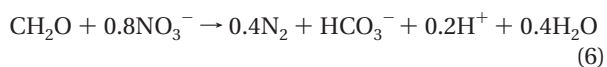
and nitrate



and for DOC mineralization using oxygen



or nitrate



or sulfate



Metal sorption reactions other than ion exchange reactions were not included in the present reaction network.

Pyrite Oxidation Kinetics. Kinetically controlled pyrite oxidation by oxygen can be modeled, e.g., following the rate expression proposed by Williamson and Rimstidt (30)

$$r_{\text{pyr}} = C_{\text{O}_2}^{0.5} C_{\text{H}^+}^{-0.11} \left(10^{-10.19} \frac{A_{\text{pyr}}}{V} \right) \left(\frac{C}{C_0} \right)_{\text{pyr}}^{0.67} \quad (8)$$

where C_{O_2} is the dissolved oxygen concentration, C_{H^+} is the proton concentration, C/C_0 is a factor that accounts for changes in surface area resulting from the progressing reaction and A_{pyr}/V is the ratio of pyrite surface area to solution volume. The rate equation was successfully applied to model observed data, for instance by Appelo et al. (15). More recently a modified form of eq 8 was presented that also incorporates oxidation by nitrate and dissolution in the absence of oxidants (12)

$$r_{\text{pyr}} = (C_{\text{O}_2}^{0.5} + f_2 C_{\text{NO}_3^-}^{-0.5}) C_{\text{H}^+}^{-0.11} \left(10^{-10.19} \frac{A_{\text{pyr}}}{V} \right) \left(\frac{C}{C_0} \right)_{\text{pyr}}^{0.67} + ((1 - \Omega_{\text{pyr}})/51) \quad (9)$$

where f_2 is a constant, which was assumed to be unity here, as in ref 12, $C_{\text{NO}_3^-}$ is the concentration of nitrate and Ω_{pyr} is the saturation ratio of pyrite. The $(1 - \Omega_{\text{pyr}})/51$ term was introduced to allow precipitation or dissolution in the absence of oxidants if the aqueous solution is supersaturated or undersaturated, respectively. For the present study this term was not considered, but eq 9 was further modified and an Arrhenius equation based temperature factor f_{ar} was incorporated. The Arrhenius equation has been used previously for comparable reactions to incorporate temperature dependency into rate equations for the bio-oxidation of ferrous

TABLE 1. Initial (Ambient) and Injectant Water Composition (Range, Typical Winter Water Composition, Typical Summer Water Composition) Used for the Model Simulations

aqueous component	initial concns (mol L ⁻¹) ^c	injectant concns (range) (mol L ⁻¹) ^c	injectant concns (21/01/1997) (mol L ⁻¹) ^c	injectant concns (05/08/1997) (mol L ⁻¹) ^c
pH	6.68	6.75–7.35	7.1	6.9
pe	–2.45	n.a.	n.a.	n.a.
Na	6.75×10^{-4}	$0.87\text{--}2.26 \times 10^{-3}$	1.26×10^{-3}	1.26×10^{-3}
Cl	2.54×10^{-4}	$1.56\text{--}3.24 \times 10^{-3}$	1.82×10^{-3}	2.04×10^{-3}
K	1.34×10^{-4}	$1.02\text{--}1.99 \times 10^{-4}$	1.23×10^{-4}	1.23×10^{-4}
Ca	2.06×10^{-3}	$1.2\text{--}2.0 \times 10^{-3}$	1.87×10^{-3}	1.42×10^{-3}
Mg	5.88×10^{-4}	$2.39\text{--}4.53 \times 10^{-4}$	3.25×10^{-4}	2.96×10^{-4}
O(0)	0	$3.5\text{--}8.13 \times 10^{-4}$	8.13×10^{-4}	3.69×10^{-4}
N(5) ^a	0	$2.42\text{--}5.48 \times 10^{-4}$	3.54×10^{-4}	2.42×10^{-4}
N(3) ^a	0	0	0	0
N(0) ^a	0	0	0	0
Amm ^b	0	0	0	0
S(6) ^a	5.52×10^{-5}	$3.54\text{--}8.96 \times 10^{-4}$	4.90×10^{-4}	4.69×10^{-4}
S(–2) ^a	0	0	0	0
Fe(2) ^a	1.04×10^{-4}	0	0	0
Fe(3) ^a	1.34×10^{-12}	3.58×10^{-8}	2.68×10^{-7}	5.37×10^{-7}
Si	2.81×10^{-4}	$0.18\text{--}3.03 \times 10^{-4}$	3.03×10^{-4}	1.35×10^{-4}
C(4) ^a	8.45×10^{-3}	$1.67\text{--}3.14 \times 10^{-3}$	3.14×10^{-3}	2.00×10^{-3}
C(–4) ^a	0	0	0	0
DOC	0	1×10^{-4}	1×10^{-4}	1×10^{-4}
temp	17 °C	1.9–22.8 °C.	1.9 °C	21.5 °C.

^a Values in brackets indicate valence. ^b Ammonium/ammonia (valence = –3). ^c Except temperature.

iron (31) and for the oxidation of chalcopyrite (32). The rate for the temperature-dependent oxidation of pyrite is

$$r_{\text{pyr}} = \frac{f_{\text{ar}}(T_c)}{f_{\text{ar}}(T_{\text{ref}})} \left[(C_{\text{O}_2})^{0.5} + f_2 C_{\text{NO}_3^-}^{0.5} C_{\text{H}^+}^{-0.11} \left(10^{-10.19} \frac{A_{\text{pyr}}}{V} \right) \left(\frac{C}{C_0} \right)_{\text{pyr}}^{0.67} \right] \quad (10)$$

with

$$f_{\text{ar}}(T_c) = -\exp\left(\frac{1}{T_c + 273.15} a_1 + a_2\right) \quad (11)$$

where a_1 and a_2 are constants, T_c is the groundwater temperature in °C and T_{ref} is a reference temperature (for example the average temperature). The temperature factor was determined for the present study by analyzing (fitting a_1 and a_2) the rate dependency of the oxygen removal rate between the injection well and port 3 of the observation well WP3, which is located 8 m away from the injection well. Note, that the approach yields a mathematical form that mimics the Arrhenius equation, but the constants a_1 and a_2 are indeed only empirical fitting parameters and cannot be compared directly to the reaction mechanism and corresponding activation energy.

Kinetics of DOC Mineralization. In many aquifers DOC acts as a competing reductant for pyrite because DOC mineralization and pyrite oxidation can occur at the same time-scale (11). In this study, DOC mineralization was modeled as a kinetic, additive Monod-term type reaction following the rate expression proposed by Parkhurst and Appelo (21), modified with the same temperature-dependency as included in eq 10

$$r_{\text{DOC}} = \frac{f_{\text{ar}}(T_c)}{f_{\text{ar}}(T_{\text{ref}})} \left[k_{\text{O}_2} \frac{C_{\text{O}_2}}{2.94 \times 10^{-4} + C_{\text{O}_2}} + k_{\text{NO}_3^-} \frac{C_{\text{NO}_3^-}}{1.55 \times 10^{-4} + C_{\text{NO}_3^-}} + k_{\text{SO}_4^{2-}} \frac{C_{\text{SO}_4^{2-}}}{1.0 \times 10^{-4} + C_{\text{SO}_4^{2-}}} \right] \quad (12)$$

where k_{O_2} , $k_{\text{NO}_3^-}$ and $k_{\text{SO}_4^{2-}}$ are rate constants with respect to dissolved oxygen, nitrate and sulfate, respectively, and $C_{\text{SO}_4^{2-}}$ is the concentration of sulfate in the groundwater. Following a two-step approach (25, 33–5), that is, a partial equilibrium approach (PEA) where the rate-limiting (kinetically controlled) electron donating oxidation step is separated from the (instantaneous) electron accepting step, the geochemical response of the DOC mineralization can be modeled within the PHT3D/PHREEQC-2 framework via addition (titration) of CH_2O at the same rate (r_{DOC}) at which DOC is removed. Using this approach electron acceptors will be automatically consumed in the order of their thermodynamic favorability, and the electron accepting reactions do not need to be defined explicitly as kinetic reactions. All reactions, whether equilibrium or kinetically controlled, are solved simultaneously by PHREEQC-2.

Modeled Ambient Groundwater Composition. Stuyfzand (28) reported the composition of the ambient groundwater before the start of the deep well injection experiment. Those measurements show that the hydrochemistry of the ambient water is fairly homogeneous within the aquifer, despite the occurrence of distinct layers that contain different concentrations of pyrite, sediment-bound organic carbon (BOM) and have different cation exchange capacities. Thus, in the numerical model homogeneous initial concentrations were assumed for the aqueous solution, based on the hydrochemistry of one representative background sample. This measured water composition was charge-balanced with PHREEQC-2 by adjusting the chloride concentration. The (heterogeneous) initial cation concentrations on the exchanger site(s) were subsequently determined for five distinct zones (16). The initial and injectant water composition used in the simulations as well as the initial exchanger composition and the mineral concentrations are listed in Tables 1 and 2, respectively.

Modeled Injectant Composition. The injected water was sourced from the Zuid-Willemsvaart canal (> 70% River Meuse water) and subsequently pretreated to remove heavy metals and pesticides that occur in trace concentrations. The hydrochemical composition of the injectant differs clearly from the ambient groundwater and generally has a more oxidized character. Furthermore, the hydrochemical com-

TABLE 2. Equilibrated Initial Exchanger Compositions, Initial Mineral Concentrations and Initial Bulk Organic Matter (BOM) Concentrations for Zones A–E^a

cations	Zone A (mol L ⁻¹)	Zone B (mol L ⁻¹)	Zone C (mol L ⁻¹)	Zone D (mol L ⁻¹)	Zone E (mol L ⁻¹)
Ca-X2	1.19×10^{-1}	2.14×10^{-2}	7.36×10^{-2}	1.89×10^{-2}	5.92×10^{-2}
Mg-X2	2.13×10^{-2}	3.84×10^{-3}	1.32×10^{-2}	3.39×10^{-3}	1.06×10^{-2}
Na-X	1.60×10^{-3}	2.87×10^{-4}	9.89×10^{-4}	2.54×10^{-4}	7.95×10^{-4}
K-X	1.57×10^{-3}	2.83×10^{-4}	9.72×10^{-4}	2.50×10^{-4}	7.82×10^{-4}
Fe-X2	2.45×10^{-3}	4.41×10^{-4}	1.52×10^{-3}	3.90×10^{-4}	1.22×10^{-3}
minerals and BOM	mol kg ⁻¹ dw	mmol kg ⁻¹ dw	mmol kg ⁻¹ dw	mmol kg ⁻¹ dw	mmol kg ⁻¹ dw
CaCO ₃	0	0	0	0	0
Fe(OH) ₃ (a)	0	0	0	0	0
pyrite	34	12	54	7	71
BOM as C	417	67	250	33	333

^a Zonation after ref 28 and as shown in Figure 1.**TABLE 3. Parameter Values Used in the Calibrated Model**

parameter	eq	value	source for value
R _T	2	2.1	estimated from eq 2
$k_{\text{pyr}} = A_{\text{pyr}} / (V \times C_{\text{pyr}})^a$	10	$115 \text{ dm}^{-1} \text{ mol}^{-1} \text{ l}$	model calibration
f_2	10	1	(12)
a_1	11	-6758.1	data analysis, as described
a_2	11	16.1	data analysis, as described
k_{O_2}	12	$1.57 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$	(20), model calibration
k_{NO_3}	12	$1.67 \times 10^{-11} \text{ mol L}^{-1} \text{ s}^{-1}$	(20), model calibration
$k_{\text{SO}_4^{2-}}$	12	$1.0 \times 10^{-13} \text{ mol L}^{-1} \text{ s}^{-1}$	(21)

^a A_{pyr}/V was computed in the numerical model from $C_{\text{pyr}} \times k_{\text{pyr}}$, where C_{pyr} is the concentration of pyrite.

position of the injectant varied significantly during the trial period, generally following a seasonal pattern. For the reactive model the same temporal discretization was used as for the preceding chloride and heat transport simulations. The measured hydrochemical composition was equilibrated and charge balanced through the modification of the chloride concentration for all 39 stress-periods. The resulting chloride concentrations in most cases differed only slightly (<5%) from the measured concentrations.

Model Calibration of the Reactive Transport Model. As the selected reaction network for the Dizon site is mainly based on equilibrium reactions, the model includes only a few adjustable parameters, i.e., the rate-determining parameters of the two kinetic reactions described by eqs 10 and 12. The model calibration focused therefore mainly on estimating the magnitude and the relative importance of those two reactions. Matching simulated and measured sulfate concentrations served as the main constraint for the estimation of the parameters that determine pyrite oxidation rates, while matching simulated and observed oxygen and nitrate concentrations served as the main constraint for the combined effect of pyrite oxidation and DOC degradation. The calibration procedure indicated that pyrite oxidation alone would account for a large fraction of the observed oxygen and nitrate removal. Therefore the modeling results were less sensitive with respect to the rate constants used in eq 12, and the original values proposed in ref 21 were employed in the final, calibrated model. Under different circumstances the agreement between simulated and measured TIC concentrations could also have served as an important constraint for the identification of DOC mineralization rates, but given the lower importance of this process and in the presence of elevated ambient TIC concentrations, this was not the case here. Model runs with and without cation exchange reactions indicated a minor impact of ion-exchange reactions on the match between simulated and observed breakthrough curves of cations. The parameter values used in the calibrated model are listed in Table 3.

Results and Discussion

Chloride and Heat Transport. Only small changes needed to be made to the originally proposed hydraulic conductivity distribution (28) to get a satisfactory fit of the chloride breakthrough curves. Figure 2 shows the modeled chloride breakthrough for the calibrated model in comparison with observed chloride concentrations. The calibrated model is furthermore capable of reproducing the observed temperature variations at most locations within the model domain. However, there were a few locations where simulation results and observations did not agree well. Those locations appear to be the ones that are located in zones of lower hydraulic conductivity and more pronounced temperature gradients, that is, the zones where conductive heat transport can be expected to play a more important role. This conclusion is supported by the fact that chloride concentrations at those locations were modeled reasonably well. For example, the observed (and modeled) chloride concentrations at the observation ports in WP1–F1, WP2–F1, and WP3–F1, respectively, remain at the level of the ambient groundwater, whereas the temperature recordings for those locations exhibit the influence of the deep well injection. An animation of the chloride transport simulation is available online (chloride.avi) and as Supporting Information.

Temperature-Dependence of Pyrite Oxidation and DOC Mineralization. The measured oxygen concentrations at port F3 of observation well WP3 varied significantly during the injection experiment. At port F3 the highest concentrations correspond to periods when the temperature of the injection water was lowest, while no breakthrough of oxygen occurred when the injectant temperatures rose above (approximately) 14 °C. The measured differences (for 2 °C < T < 14 °C) between injection well and WP3–F3 were used to determine the unknown coefficients in eq 11 (see Figure 3, values in Table 3). Unfortunately the analysis does not allow for the differentiation of the temperature-dependencies of the pyrite oxidation and the DOC mineralization rates. After incorpo-

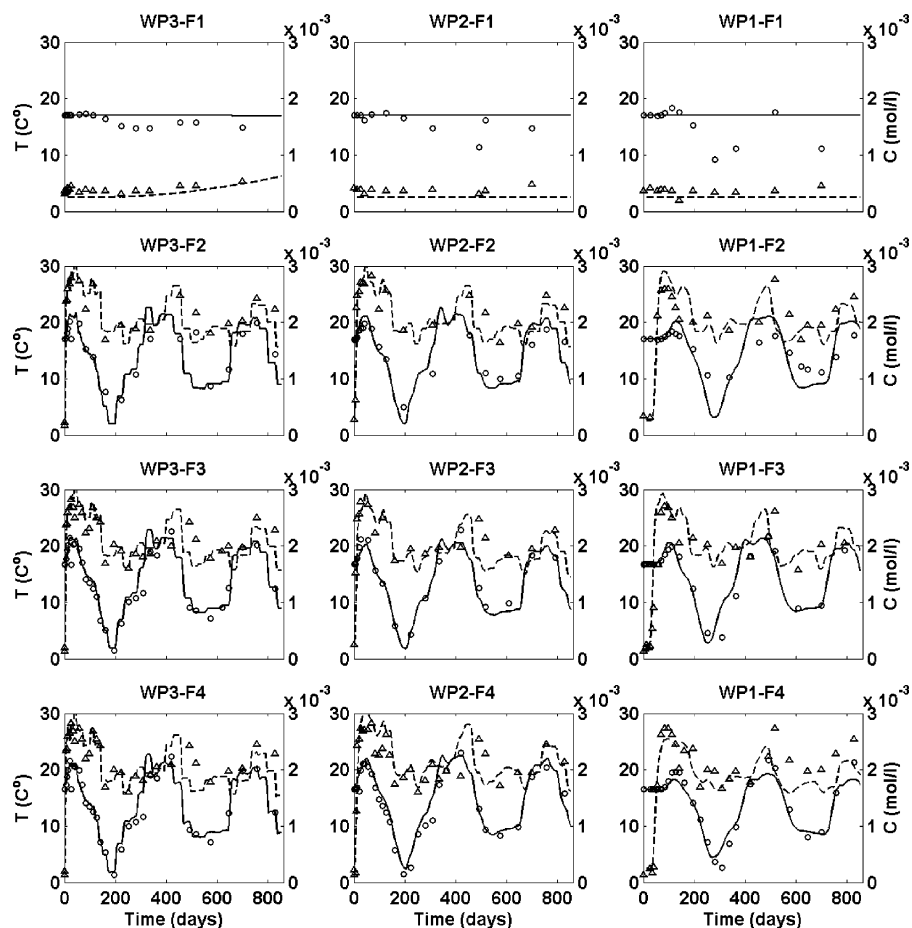


FIGURE 2. Measured and simulated concentrations of chloride and temperature for WP1 (38m), WP2 (12m) and WP3 (8m). Simulated concentrations are indicated by solid (Cl^-) and dashed (temperature) curves, respectively. Triangles (Δ) indicate measured chloride concentrations, circles ($\circ$) indicate measured temperatures.

rating the temperature-terms into eqs 10 and 12 the calibrated reactive transport model reproduces the observed oxygen removal very well. Moreover, the even lower breakthrough concentrations at WP2-F3 (same port/depth) and the lack of breakthrough at WP1-F3 is also well described by the model, as is the behavior of nitrate. Both, measured and simulated nitrate concentrations at WP3-F3, WP2-F3 and WP1-F3, all located in Zone D (Figure 1), show a clear seasonal variation. However, the comparison between reactive and nonreactive model runs (data not shown) indicates that the breakthrough curves of nitrate at WP3-F3 and WP2-F3 are not significantly affected by reactions. In contrast, the same comparison for the simulated breakthrough curve at WP1-F3 shows that the breakthrough of nitrate is substantially affected by reactive processes, leading to complete nitrate removal for some short periods during which the injectant temperature was highest. In Figure 4 the measured and the simulated concentrations of oxygen and nitrate are shown for WP3-F3, WP2-F3 and WP1-F3 in comparison with results from three model runs for which temperatures were fixed in time and space at 8, 14 and 20 °C, respectively. The comparison demonstrates that the observations were best matched by the model run which used the simulated temperature field for the computation of the reaction rates. During the 854 days simulation time approximately 9.11×10^6 mol electron equivalents (Eeq) were added to the aquifer via deepwell injection, 5.45×10^6 mol as nitrate and 3.65×10^6 mol as oxygen. The analysis of the model results (Figure S3, Table S1) indicates that the largest part of the added oxidation capacity was consumed by pyrite oxidation (6.34×10^6 mol Eeq), while only 1.64×10^6 mol Eeq were consumed by BOM mineralization.

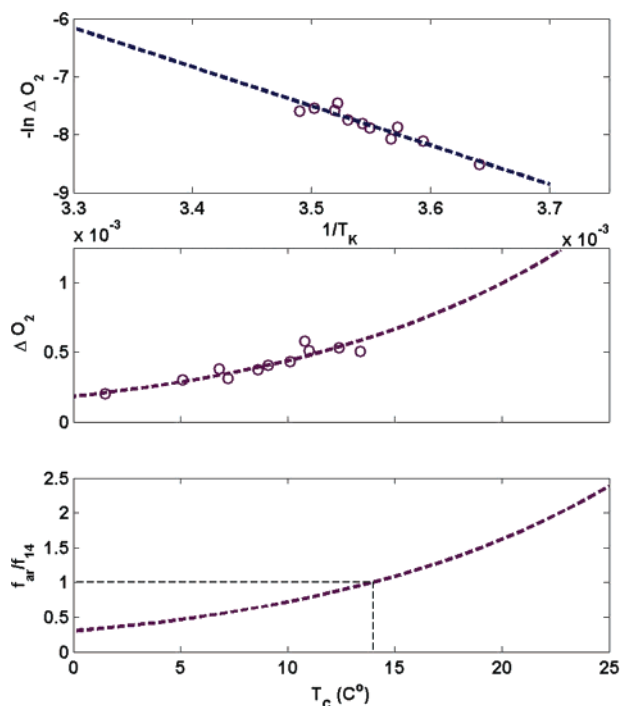


FIGURE 3. Measured and fitted oxygen removal rate between the injection well and observation point WP3.3 (first and second row) and the resulting normalized (with respect to $T_c = 14$ °C) temperature factor f_d/f_{14} .

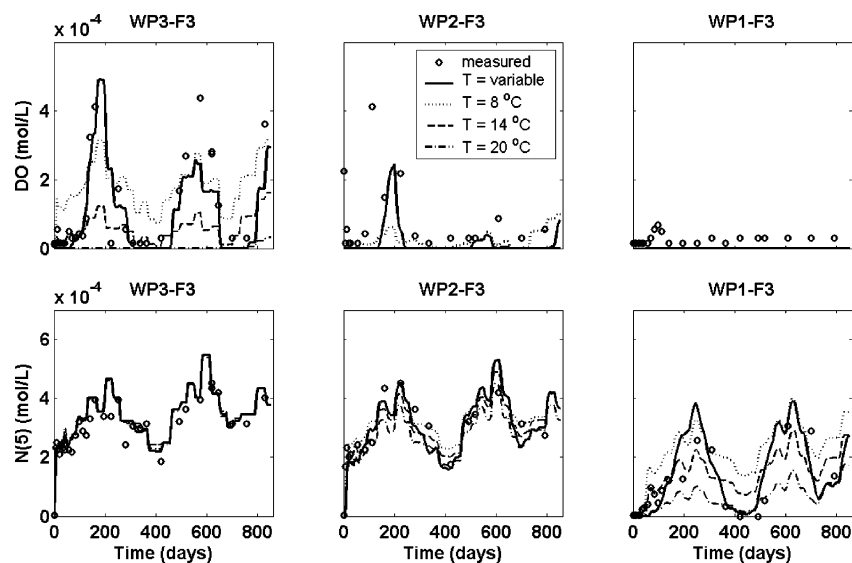


FIGURE 4. Measured and simulated breakthrough curves of oxygen and nitrate at three different distances from the injection well (WP3 = 8m, WP2 = 12m, WP1 = 38m), in aquifer zone D (F3). Simulated concentrations are shown for three cases where the temperature T was fixed in space and time (at 8, 14 and 20 °C, respectively) in comparison with the case where the simulated temperature field (T variable in space and time) was used.

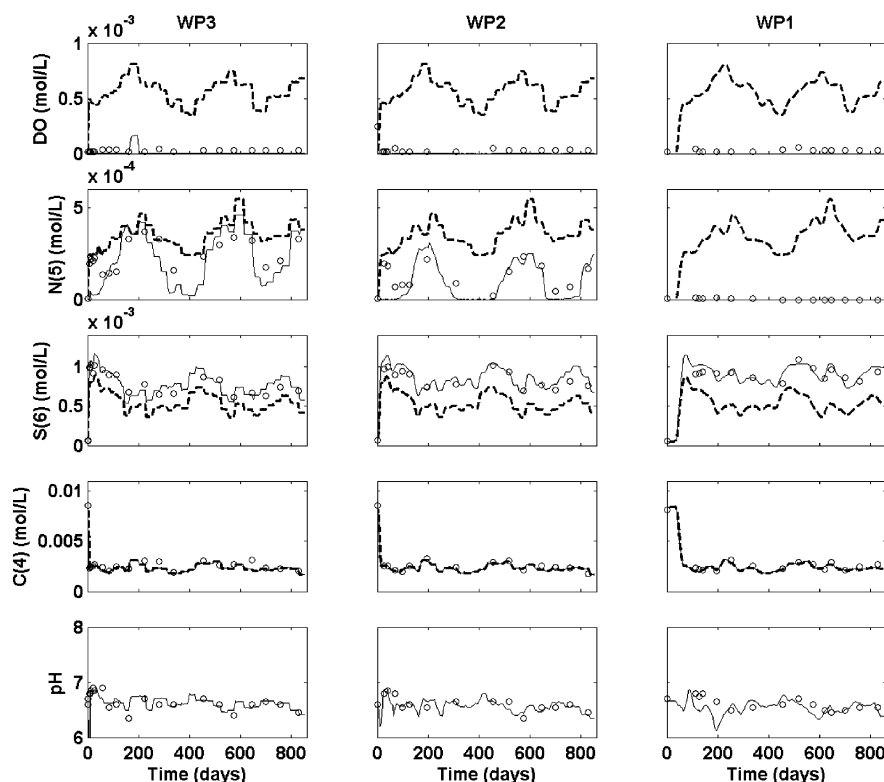


FIGURE 5. Simulated (solid lines) and measured (circles) aqueous component concentrations in "Zone E". For comparison, the dashed lines indicate the simulation results for advective-dispersive transport (no reaction).

Influence of Aquifer Heterogeneity. The locations and rates at which oxygen and nitrate are removed from the injectant can vary significantly as a result of physical and chemical heterogeneity of the aquifer, for example due to the variability of the hydraulic conductivity and differences in the pyrite content of the aquifer matrix. The effect can be seen for the measured and simulated breakthrough curves at WP3–F2, WP2–F2 and WP1–F2 (Figure 5), which show the water quality changes in Zone E (see Figure 1), at similar locations as WP3–F3, WP2–F3 and WP1–F3, but at different depth. Breakthrough of oxygen was measured in none of those locations, which was well reproduced by the model.

Compared to Zone D (Figure 4) nitrate concentrations are generally lower, except for WP3–F2 during the first winter period (at ~200 days after the start of the experiment) when essentially no or little nitrate was removed as a result of the low temperature of the injectant. In contrast to Zone D, which has a very low pyrite content, no breakthrough of nitrate occurred at WP1, located 38 m from the injection well.

Reactive vs Nonreactive Transport. The significant impact of the injectant temperature on the variability of nitrate removal rates becomes additionally evident from the comparison between the reactive simulation with the calibrated model and a corresponding model run where

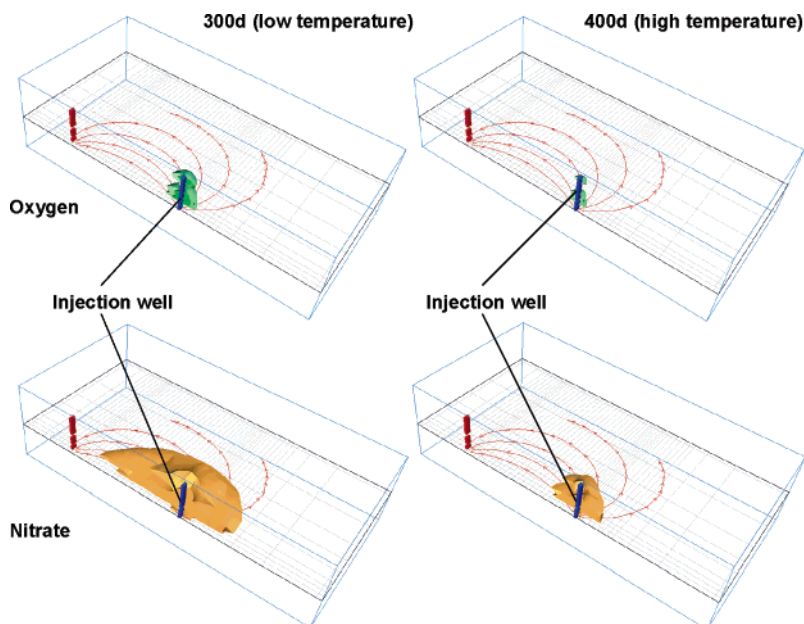


FIGURE 6. Illustration of the seasonally variable redox-zonation — simulated isoconcentration surfaces of oxygen (top row) and nitrate (bottom row) at 300 days (left) and at 400 days (right) after the start of the deep well injection experiment. Also shown are the projected pathlines between injection and extraction well (arrows indicate 100 days travel time) in “Zone E”.

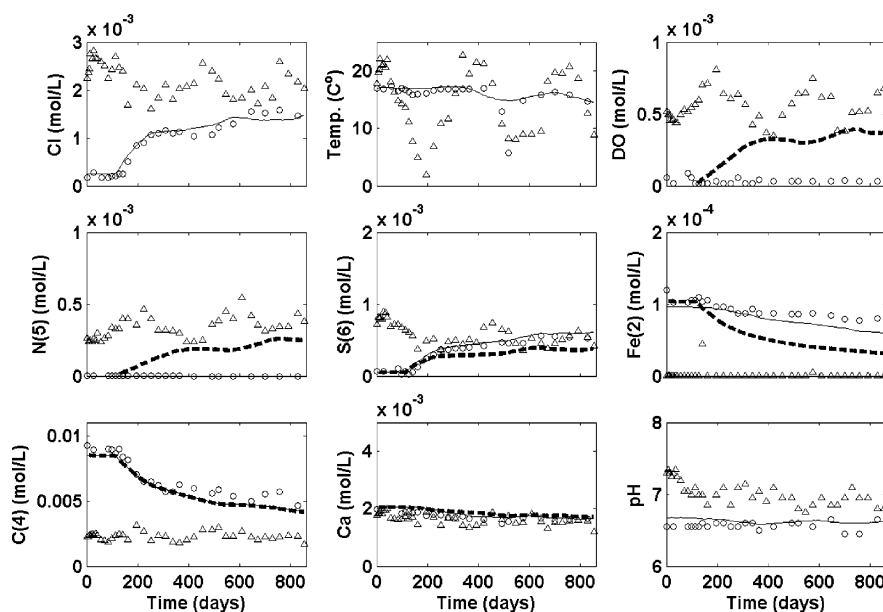


FIGURE 7. Selected simulated (solid lines) and measured (circles) aqueous component concentrations at the extraction well PP.1 in comparison with measured concentrations at the injection well IP.2 (triangles). The dashed lines indicate the simulated extraction well concentrations for advective-dispersive transport (no reaction).

reactions were deactivated. As can be seen in Figure 5, the differences between modeled nitrate concentrations at WP3–F2 differs strongly at times, following a seasonal pattern. The comparison of the two model runs allows for distinguishing the patterns that result from the seasonal variability of nitrate in the injectant, and the changes that result from changes in reaction rates, for example in denitrification rates. The comparison indicates also to what extent sulfate concentrations are impacted by pyrite oxidation. As Figure 5 shows for Zone E, the measured and simulated sulfate concentrations increase with travel distance and are at WP1–F2 almost twice as high as the sulfate concentrations predicted by the nonreactive model. No significant discrepancies occurred between reactive and nonreactive model runs for the total inorganic carbon (TIC) concentrations, indicating that DOC mineralization has a minor impact on the TIC concentrations.

Large-Scale Temporal Changes of the Redox Zonation.

Comparison of the simulations with measured data at further observation points show that the calibrated model provides, for most parts of the domain, an excellent qualitative and quantitative description of the coupled transport and major reactions that occurred during the deep well injection experiment, including pH and pe (redox state). Because the latter two are the key variables for many geochemical and biological processes such as for the mobility of trace metals and for the attenuation of organic micropollutants, the present model provides a good foundation for a more detailed quantification framework that specifically deals with those aspects in artificially recharged systems. The good agreement between simulation and observations is thought to result mainly from the careful initial hydrogeological and hydrochemical characterization of the aquifer in combination with

the incorporation of the temperature-dependency of the pyrite oxidation reaction. Figure 6 illustrates in three dimensions the simulated variability of the redox zonation for two selected times, one representing “low” temperature conditions (after 300 days simulation time) and the second (after 400 days simulation time) representing “high” temperature conditions. The comparison of the concentration isosurfaces for the two times shows how the oxygen and nitrate “bubbles” shrink significantly within 100 days. Movies (animations) for the full simulation period are available online (oxygen.avi, nitrate.avi) and as Supporting Information.

Extraction Water Quality. The quality of the extracted water is a mixture of the ambient water and of a steadily increasing fraction of the injected water, modified by the previously discussed reactions. Figure 7 shows the observed and the simulated breakthrough curves of the major reactants in comparison with the corresponding concentrations of the injectant and in comparison with the simulation results for the case of nonreactive transport. The initial breakthrough of water quality changes, for example for chloride, exhibits a significant dispersion, despite the small longitudinal ($\alpha_L = 1$ cm) and transversal (local) dispersivities ($\alpha_T = 1$ mm) that were used in the model simulations. The dispersion seen in those breakthrough curves at the extraction well is largely created through the (i) conductivity differences between layers in combination with (ii) the specific flow-field in x–y direction that creates a broad spectrum of travel times between the injection and the extraction well. The latter point is illustrated in Figure 6, where different pathlines (with indication of travel times) between the injection and the extraction well are shown for a layer corresponding to “Zone E”. The dispersivity of the flow field leads to an almost complete dissipation of the seasonal variations of the injectant water quality and the seasonality created by the temperature-dependent variable reactivity. The comparison of the corresponding sulfate and dissolved iron concentrations from the reactive and from the nonreactive model permits an identification and quantification of the water quality changes that result from pyrite oxidation. Again, the impact of DOC mineralization on TIC concentrations appears to be negligible.

Acknowledgments

The present work was partially supported by the Centre for Groundwater Studies Australia, AWWA Research Foundation and CORONA, a research project supported by the European Commission under the Fifth Framework Program, Key Action 1 ‘Sustainable Management and Quality of Water’, within the thematic program for Energy, Environment and Sustainable Development. The water supply companies Brabant Water and WML are acknowledged for their permission to use the experimental data.

Supporting Information Available

An additional table, three figures, and three animations showing the simulated spatial-temporal evolution of the three-dimensional concentration isosurfaces of chloride, oxygen and nitrate, respectively. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Received for review August 24, 2004. Revised manuscript received January 13, 2005. Accepted January 14, 2005.

ES0486768